

ATTICUS MUSONI

FRIDAY LEARNING PACKET

TODAY'S BIG GOAL

Explain what you know - not just produce an answer.

Today closes the first school week. You will use Thursday feedback, learn one new Grade 6 math skill, strengthen academic vocabulary, and finish with an online AI experiment and a portfolio-ready design piece.

YOUR THREE RULES TODAY

1. Read the glossary before every subject.
2. Listen to the lesson explanation before independent work.
3. When a question says EXPLAIN, give the reason - not only the result.

Day 4 of 180 | 176 instructional days remain after today

Friday Schedule

Time	Class / Mission
8:30-8:50	Thursday feedback + Week 1 review
8:50-9:50	Mathematics 6 - Decimal Division
9:50-10:05	Break
10:05-10:55	Literature - Central Idea + Objective Summary
10:55-11:35	Writing - Explanatory Paragraph
11:35-12:15	Lunch / outside break
12:15-1:05	Science - Transparent, Translucent, Opaque
1:05-1:50	History & Geography - From Geography to the Polis
1:50-2:05	Break
2:05-2:35	French 6 - Negation, Elision, -ER Accuracy
2:35-3:25	Computer Science / AI - Rules vs Machine Learning
3:25-4:00	Art / Design - Week 1 Visual Explainer
4:00-4:10	Weekly closeout - Nothing Missing

WHAT THURSDAY PROVED

- Fraction division improved dramatically: your visible independent and exit-ticket computations were correct.
- Decimal multiplication is mostly strong, but one exit-ticket error shows why estimating before calculating must become automatic.
- Reading comprehension is solid. Your next step is making the TOPIC more specific and writing a complete OBJECTIVE SUMMARY.
- Science: your light-source -> object -> eye model is strong. Today we will tighten the words transparent, translucent, and opaque.
- History: you correctly connected mountains to separate city-states. Today we fix the role of the sea and the difference between cultural connection and political unity.
- French: ne...pas is mostly secure. Today we fix elision (n'...) and subject/verb endings.
- Computer Science/AI: you successfully completed Scratch and Teachable Machine. Today you will explain the difference between a programmed rule and learned patterns.

Mathematics

WHAT YOU DID WELL

Fraction division is now much stronger. You correctly solved $5/6 \div 1/3 = 2 \frac{1}{2}$, $7/10 \div 7/20 = 2$, $2 \frac{1}{4} \div 3/8 = 6$, and $4 \frac{2}{3} \div 1 \frac{1}{6} = 4$.

Your fraction-division exit items were also correct: $3/5 \div 9/10 = 2/3$ and $1 \frac{1}{2} \div 3/4 = 2$.

Most decimal multiplication was accurate: $14.8 \times 0.35 = 5.18$; $0.84 \times 2.75 = 2.31$; $6.42 \times 1.8 = 11.556$; $12.05 \times 0.6 = 7.23$; $0.375 \times 8 = 3$; and $\$12.75 \times 4 = \51 .

WHAT TO IMPROVE

On 7.2×0.45 , you wrote 5.24. The correct answer is 3.24. An estimate of $7 \times 0.5 \approx 3.5$ would have warned you that 5.24 was too large.

Your explanation of why $8 \div 1/2$ is greater than 8 gave the result (16), but not the idea. Dividing by $1/2$ asks: "How many halves fit into 8?" There are 16 halves.

2-minute correction. Estimate, then correct: $7.2 \times 0.45 =$

Hint: Use 7×0.5 as a quick reasonableness check.

Literature / Reading

WHAT YOU DID WELL

You correctly explained the asphalt cause/effect chain: sunlight is absorbed, the surface warms, and nearby air can warm.

You correctly defined transpiration from context as plants releasing water from leaves into the air.

You identified two cooling effects of vegetation: shade and water release.

Your evidence about shaded/grassy areas versus asphalt/buildings was relevant to the idea of small local climate differences.

WHAT TO IMPROVE

"Difference" is too broad to be a useful topic. A stronger topic would be "temperature differences within cities" or "urban heat."

Your objective summary was too short to capture the main causes. A complete summary should include the phenomenon plus the major reasons for it.

When a question asks why repeated measurements matter, "one is not enough" is the start. Add the reason: repeated measurements reduce the chance that one unusual reading gives a misleading conclusion.

8:30-8:50

THURSDAY FEEDBACK - SCIENCE + HUMANITIES + TECH

Use these corrections during today's lessons.

Science

STRONG CONCEPT

Your light-path explanation was strong: room light -> book -> reflected light -> eyes. Your dark-room challenge was also correct.

VOCABULARY REPAIR

Transparent: passes most visible light with little scattering; translucent: passes some light but scatters it; opaque: does not transmit visible light.

Key idea: light can pass through a material even when you cannot see a clear image through it.

History & Geography

WHAT YOU DID WELL

Your map was strong, and you correctly connected mountains -> difficult travel -> independent city-states.

ONE IMPORTANT CORRECTION

The sea mattered for travel, trade, fishing, communication, and movement of goods/ideas - not as a drinking-water source.

Today we finish the difference between cultural connection and political unity.

French + Computer Science / AI

FRENCH

French: ne...pas is mostly secure. Repair elision (elle n'écoute pas) and subject endings (je joue / tu joues).

CS/AI: both online builds were completed. Today we finish the difference between programmer-written rules and patterns learned from labeled examples.

WORDS YOU NEED TODAY

quotient - the answer to a division problem

dividend - the number being divided

divisor - the number you divide by

equivalent - equal in value even if written differently

estimate - a close prediction used to check whether an exact answer makes sense

magnitude - the size of a number or quantity

LISTEN FIRST - VOICE TUTOR PROMPT

You are teaching Atticus Grade 6 decimal division. Begin by praising the specific improvement in fraction division without overpraising. Remind him that Thursday showed strong fraction-division procedure and mostly strong decimal multiplication, with one decimal-product error that estimation could have caught. State today's objective: divide decimals accurately, make a decimal divisor into a whole number by multiplying BOTH dividend and divisor by the same power of 10, and use estimation to check the quotient. Teach the meaning of dividend, divisor, quotient, equivalent, estimate, and magnitude in plain language. Work through three complete examples before independent work: $7.2 \div 3$, $8.64 \div 0.27$, and $4.5 \div 0.75$. For each, estimate first and explain why changing both numbers by the same factor keeps the quotient equal. Ask a verbal check: "In $6.3 \div 0.9$, which number must become whole, and what do we multiply both numbers by?" Wait for Atticus. Use hints before answers. Do not reveal exit-ticket answers.

Today's learning target

By the end of this lesson, you should be able to divide decimals by whole numbers and by decimals, explain why BOTH numbers move together when the divisor is a decimal, and use an estimate to check the size of your answer.

FIRST: FIX THURSDAY'S ONE DECIMAL ERROR

7.2×0.45 should be close to $7 \times 0.5 = 3.5$. Exact calculation: $72 \times 45 = 3240$, with three decimal places total $\rightarrow 3.240 = 3.24$.

How to divide decimals

CASE 1 - The divisor is already a whole number: divide normally and keep place value in mind.

CASE 2 - The divisor is a decimal: make the divisor a whole number. Multiply the divisor AND the dividend by the same power of 10. This creates an equivalent division problem.

Always estimate before or after calculating. If your exact quotient is wildly different from the estimate, inspect your decimal placement.

WORKED EXAMPLE 1: Decimal divided by a whole number

Problem: $7.2 \div 3$

Solution: 2.4

Why it works: Think of 72 tenths $\div 3 = 24$ tenths = 2.4. Estimate: $6 \div 3 = 2$, so 2.4 is reasonable.

WORKED EXAMPLE 2: Decimal divisor

Problem: $8.64 \div 0.27$

Solution: 32

Why it works: The divisor 0.27 has two decimal places. Multiply BOTH numbers by 100: $864 \div 27 = 32$. The quotient stays the same because both numbers were scaled by the same factor.

WORKED EXAMPLE 3: Application

Problem: 4.5 liters $\div$ 0.75 liter per bottle

Solution: 6 bottles

Why it works: Multiply both by 100: $450 \div 75 = 6$. Check: 0.75 is about $\frac{3}{4}$, and six groups of $\frac{3}{4}$ make 4.5.

COMMON ERROR

Do NOT move only the decimal in the divisor. If you multiply the divisor by 10 or 100, you must multiply the dividend by the same amount.

1. $9.6 \div 4 =$

Hint: Think of 96 tenths divided by 4.

2. $6.3 \div 0.9 =$

Hint: Make 0.9 into 9 by multiplying BOTH numbers by 10.

3. $5.04 \div 0.12 =$

Hint: Multiply BOTH numbers by 100.

4. $3.375 \div 0.75 =$

Hint: Estimate first: about $3.4 \div 0.75$.

5. A 12.6-meter roll is cut into pieces that are 0.9 meter long. How many pieces?

VERBAL CHECK

Without writing: In $2.4 \div 0.06$, why would multiplying both numbers by 100 help? What equivalent division problem would you get?

Independent practice

1. $8.4 \div 0.7 =$

2. $7.56 \div 0.21 =$

3. $12.48 \div 0.24 =$

4. $2.52 \div 0.6 =$

5. A 5.4-kilogram container is divided equally into packages of 0.45 kilogram. How many packages are made?

6. A museum charges \$18.90 for 3 identical student tickets. What is the price of one ticket?

Exit Ticket - do without help

E1. $4.2 \div 0.6 =$

E2. $9.45 \div 0.35 =$

E3. $6.25 \div 0.25 =$

E4. Explain why $8.64 \div 0.27$ and $864 \div 27$ have the same quotient.

E5. Estimate $3.6 \div 0.4$, then give the exact quotient.

WORDS YOU NEED TODAY

thermal expansion - the increase in size of a material as it warms

contract - to become smaller

expansion joint - a designed gap or flexible connection that allows parts of a structure to move

stress - internal force created when a material is pushed, pulled, bent, or prevented from moving

central idea - the main point an author communicates about a topic

objective summary - a short account of the most important ideas without personal opinion

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus how to identify a precise topic, central idea, and objective summary using today's informational text about bridges. Begin by reminding him that Thursday comprehension was solid, but "difference" was too vague as a topic and his summary left out major causes. Define each glossary term in sixth-grade language. Before he reads, give one unrelated example: topic "sleep"; central idea "Regular sleep helps the brain and body function well." Explain that an objective summary should include the phenomenon/problem, the major explanation, and the result or solution - without opinion. Do NOT tell him the central idea of today's bridge article. After he reads, ask: "What is the topic in 2-5 words?" Then: "What does the author most want us to understand about that topic?" Push him to explain which details support the central idea. If he gives a vague response, ask him to make the noun/topic more specific.

READING TARGET

By the end of this lesson, you should be able to name a precise topic, state a complete central idea, explain how details support it, and write a 2-3 sentence objective summary.

Why Bridges Need Room to Move

A bridge may look completely still, but its materials are always responding to the environment. Steel, concrete, and other building materials can change size slightly as their temperature changes. When many meters of material are connected together, tiny changes in each section can add up to noticeable movement across the whole structure.

When a solid material warms, its particles move more vigorously. In many materials, that increased motion causes the material to take up a little more space. Engineers call this thermal expansion. When the material cools, it usually contracts. A short metal ruler may change by such a small amount that you would never notice it, but a long bridge can expand or contract by several centimeters over a large temperature range.

The movement itself is not necessarily a problem. The danger comes when a structure is not allowed to move. Imagine pushing a long object against two rigid walls while it tries to become slightly longer. Pressure and stress build inside the material. Over time, repeated stress can contribute to cracking, bending, or damage to connections.

To control this problem, engineers build movement into the design. Many bridges include expansion joints - gaps or flexible connections between sections. These joints allow parts of the bridge deck to move slightly while vehicles still cross safely. Some bridges also use bearings that let the deck shift a small amount over the supports beneath it.

Engineers do not simply guess how much movement will occur. They consider the materials used, the length of the bridge, the expected temperature range, and the forces the bridge must carry. A bridge in a place with very hot summers and cold winters may need to handle a larger temperature change than a bridge in a mild climate.

Expansion joints are only one example of a larger engineering idea: strong structures are often designed to move in controlled ways. Buildings can sway slightly in wind. Tall towers can flex. Bridges can expand, contract, and vibrate. Good engineering does not always mean preventing movement. Sometimes it means predicting movement and giving it a safe path.

SECOND READ - MARK THESE

- Underline one sentence that explains WHY bridge materials move.
- Circle one detail that explains what can go wrong if movement is blocked.
- Put a star beside one engineering solution.

1. What is the **TOPIC** of the article? Answer in a precise 2-5 word phrase.

2. State the **CENTRAL IDEA** as one complete sentence.

3. What happens to many bridge materials when they warm? Explain the cause and effect.

4. In context, what does “stress” mean in paragraph 3?

5. Why can blocking normal expansion cause damage?

6. Give **TWO** ways engineers allow bridges to move safely.

7. Why does the author mention a metal ruler?

8. What larger engineering idea is explained in the final paragraph?

9. Write a 2-3 sentence **OBJECTIVE** summary. Include the phenomenon, the problem, and the engineering response.

WORDS YOU NEED TODAY

explanatory writing - writing that teaches how or why something happens

topic sentence - the sentence that states the main explanation

evidence - a fact or detail that supports an explanation

reasoning - the explanation of why the evidence proves or supports the main idea

transition - a word or phrase that helps ideas connect

revise - to improve ideas, organization, wording, or sentence clarity

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus to write one strong Grade 6 explanatory paragraph using the bridge reading. Connect to Thursday: his ideas are often correct, but some written responses stop too early or are too vague. Define topic sentence, evidence, reasoning, transition, and revise. Model the difference between evidence alone ("Steel expands when heated.") and evidence plus reasoning ("Steel expands when heated, so a long bridge needs space for that movement; otherwise stress can build in the structure."). Then model a 5-sentence mini-paragraph with a topic sentence, evidence 1 + reasoning, evidence 2 + reasoning, and conclusion. Before Atticus writes, ask him verbally to state his main explanation and two pieces of evidence. Do not compose his final paragraph. Require 8-10 complete sentences and at least two facts from the article. At the end, have him read one sentence aloud and revise it if it is too long, vague, or incomplete.

Prompt

Why do engineers design bridges with expansion joints or other ways to allow movement?

PLAN BEFORE WRITING

Main explanation: _____

Evidence 1: _____

Why it matters: _____

Evidence 2: _____

Why it matters: _____

Conclusion idea: _____

WORKED EXAMPLE 1: Evidence only

Problem: Steel expands when it gets warmer.

Solution: This is a useful fact, but the reader still needs the connection.

Why it works: Evidence tells WHAT is true; reasoning explains WHY it matters to your answer.

WORKED EXAMPLE 2: Evidence + reasoning

Problem: Steel expands when it gets warmer.

Solution: Because a bridge is long, small changes across many sections can add up. Expansion joints give the structure room to move without building damaging stress.

Why it works: The explanation connects the evidence directly to the design choice.

10:55-11:35

WRITING - DRAFT + REVISION

8-10 complete sentences. Use at least two facts from the article.

WRITING - DRAFT + REVISION

8-10 complete sentences. Use at least two facts from the article.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SELF-CHECK BEFORE YOU FINISH

- I answered WHY engineers allow movement.
- I used at least TWO specific facts from the article.
- After each important fact, I explained why it matters.
- I used complete sentences and checked punctuation.
- My final sentence wraps up the explanation instead of introducing a new idea.

- ## SELF-CHECK BEFORE YOU FINISH
- I answered WHY engineers allow movement.
 - I used at least TWO specific facts from the article.
 - After each important fact, I explained why it matters.
 - I used complete sentences and checked punctuation.
 - My final sentence wraps up the explanation instead of introducing a new idea.

WORDS YOU NEED TODAY

transmit - to let light pass through

scatter - to redirect light in many directions

transparent - transmits most visible light with little scattering, allowing a clear image

translucent - transmits some visible light but scatters it, so images are blurry or unclear

opaque - does not transmit visible light through the material

absorb - to take in light energy rather than transmit or reflect it

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus the difference between transparent, translucent, and opaque using the key idea that “Can I see a clear image?” is not the same question as “Does any light pass through?” Connect directly to Thursday: he correctly understood light source -> object -> eye, but the classification vocabulary needs repair. Define transmit, scatter, transparent, translucent, opaque, and absorb. Give three models: clear glass/plastic (transparent if the image is clear), wax paper or thin tissue (usually translucent because some light passes but is scattered), cardboard (opaque because visible light does not pass through). Tell him actual household materials vary, so he must record what he observes rather than force a label. Explain that a bright patch of light behind a material is evidence of transmission even if the object is not visible. Before the lab, ask him to predict which sample will transmit light most clearly. After the lab, ask him to justify each classification with evidence.

TODAY'S QUESTION

How can a material let light pass through but still prevent you from seeing a clear image?

WORKED EXAMPLE 1: Transparent

Problem: A clear window

Solution: Light from objects passes through with relatively little scattering, so details remain visible.

Why it works: A clear image is evidence of high transmission with low scattering.

WORKED EXAMPLE 2: Translucent

Problem: Wax paper or frosted plastic

Solution: Some light reaches the other side, but it is scattered in many directions.

Why it works: You may see brightness without a clear image.

WORKED EXAMPLE 3: Opaque

Problem: Cardboard

Solution: Visible light does not pass through the cardboard.

Why it works: You may see reflected light on the front, but not transmitted light behind it.

SAFETY

Never shine a bright flashlight directly into anyone's eyes. Keep water away from electronics.

Materials

Flashlight or lamp; white paper/wall; clear plastic; wax paper or thin tissue; cardboard; optional frosted/translucent plastic.

Procedure

1. Aim the flashlight at a white wall or sheet of paper from a fixed distance.
2. Observe the brightness with NO material in the path.
3. Put the clear plastic between the flashlight and wall. Record whether light reaches the wall and whether the bright spot stays sharp.
4. Repeat with wax paper/tissue.
5. Repeat with cardboard.
6. For each material, classify it based on BOTH transmission and image/spot clarity.
7. Draw one ray model showing where light is transmitted, scattered, reflected, or absorbed.

Material	Light behind?	Spot/image clear?	Best class	Evidence
No material				
Clear plastic				
Wax paper/tissue				
Cardboard				

Ray model

1. Which tested material transmitted the most light? What evidence supports your answer?

2. Did any material transmit light without giving a clear image? Explain what that tells you about scattering.

3. Explain the difference between translucent and opaque in your own words.

4. CER: Why might wax paper let a room look brighter behind it even when you cannot clearly see an object through it?

Challenge. If a material reflects some light AND transmits some light, does that automatically make it transparent? Explain.

WORDS YOU NEED TODAY

polis - an independent Greek city-state and its surrounding territory

city-state - a city and nearby territory that function as an independent political unit

politically independent - governing itself rather than being ruled by one larger central government

culturally connected - sharing language, beliefs, customs, stories, or practices even without one government

harbor - a sheltered place where ships can anchor safely

trade - exchange of goods, resources, or services

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus how Greek geography contributed to the polis. Connect to Thursday: he correctly explained that mountains divided communities and made land travel difficult. Correct the sea misconception gently: seawater was not the main benefit for drinking; the sea mattered for travel, fishing, trade, communication, and movement of goods and ideas. Define polis, city-state, politically independent, culturally connected, harbor, and trade. Give a simple modern analogy for culturally connected but politically separate places without implying they are identical to ancient Greece. Explain that Greek communities could share language, religion, festivals, stories, and trade while still having separate governments. Ask Atticus to explain the chain: geography -> travel -> separate communities -> independent political development. Then have him read the short passage and answer independently.

From Mountains and Harbors to the Polis

Ancient Greece did not develop as one large river-valley kingdom. Mountains divided much of the mainland into smaller regions, while the coastline opened toward the Aegean and Mediterranean Seas. This combination encouraged communities to grow separately on land while still connecting by water.

Many Greek communities developed as poleis (POH-lays), the plural of polis. A polis was more than a crowded town. It included a city or central settlement, nearby farmland, and the people who belonged to the community. Each polis could make its own laws, organize its own government, and defend its own territory.

Sea travel helped reduce some of the isolation created by mountains. Harbors allowed ships to carry pottery, olive oil, wine, metals, grain, people, and ideas. Fishing also supplied food. Trade became especially important because fertile farmland was limited in many areas. Communities could obtain resources they did not produce easily themselves.

Greek poleis were often politically independent, but they were not completely disconnected from one another. Many Greeks shared versions of the same language, worshiped many of the same gods, told common myths, and participated in religious festivals and athletic contests. In other words, they could be culturally connected without being politically unified under one ruler.

That combination - local independence plus wider cultural connections - became an important feature of the Greek world. It also created competition. Poleis could trade with one another, cooperate against an outside threat, or become rivals. Later in the course, Athens and Sparta will show how two Greek city-states could share important parts of Greek culture while developing very different political and social systems.

1. What was a polis?

2. Why did mountains encourage local political independence?

3. Give THREE ways the sea helped Greek communities.

4. Why was trade especially important when farmland was limited?

5. Explain the difference between culturally connected and politically unified.

6. Complete this cause/effect chain: mountains -> difficult land travel -> _____ -> _____.

7. Why could two poleis share Greek culture but still become rivals?

Exit. In 3-4 sentences, explain how BOTH mountains and the sea shaped the Greek world.

WORDS YOU NEED TODAY

subject pronoun - a word such as je, tu, il, elle, nous, vous, ils, elles that tells who does the action

stem - the part of a regular verb left after removing -er

ending - letters added to a verb stem to match the subject

conjugate - to change a verb form to match the subject

negation - a form that makes a sentence negative, such as ne...pas

elision - dropping a vowel before another vowel sound; ne becomes n' before a vowel

LISTEN FIRST - VOICE TUTOR PROMPT

You are Atticus's continuing French tutor. Thursday showed that he understands the basic ne...pas pattern but needs elision and subject/verb ending accuracy. Start with a quick oral review of regular -ER endings: je -e, tu -es, il/elle -e, nous -ons, vous -ez, ils/elles -ent. Teach elision: before a vowel or silent h, ne becomes n', as in elle n'écoute pas and je n'aime pas. Give three spoken examples and have Atticus repeat them. Then contrast je joue with tu joues. Ask four short questions in French using aimer, regarder, parler, écouter, or jouer. Require full-sentence answers. Correct gently and let him repeat. Do not introduce advanced exceptions beyond today's goal. The written task should remain Grade 6 continuing French, not beginner vocabulary drills.

Regular -ER endings

je -e | tu -es | il/elle -e | nous -ons | vous -ez | ils/elles -ent

WORKED EXAMPLE 1: Negation

Problem: Je regarde la télévision.

Solution: Je ne regarde pas la télévision.

Why it works: ne goes before the conjugated verb and pas goes after it.

WORKED EXAMPLE 2: Elision

Problem: Elle écoute la musique.

Solution: Elle n'écoute pas la musique.

Why it works: Before a vowel sound, ne becomes n'.

WORKED EXAMPLE 3: Je vs tu

Problem: jouer

Solution: Je joue. / Tu joues.

Why it works: The subject changes the verb ending: je takes -e; tu takes -es.

A. Make each sentence negative

1. J'aime les maths.

2. Tu écoutes la radio.

3. Elle regarde un film.

4. Nous parlons français.

5. Ils jouent au tennis.

B. Conjugate the verb correctly

6. Je _____ (jouer) au basket.

7. Tu _____ (aimer) les sciences.

8. Nous _____ (regarder) un film.

9. Elle _____ (écouter) la musique.

10. Ils _____ (parler) français.

C. Answer in a complete French sentence

11. Est-ce que tu aimes les maths ?

12. Est-ce que tu écoutes de la musique ?

13. Est-ce que tu regardes des films ?

14. Est-ce que tu joues à des jeux vidéo ?

ORAL CHECK

Answer at least four spoken questions in French. If the tutor corrects one sentence, repeat the repaired sentence aloud.

WORDS YOU NEED TODAY

rule-based program - a program that follows explicit instructions written by a programmer

machine learning - a method in which a system learns patterns from examples instead of receiving every decision rule directly

training data - examples used to teach a machine-learning model

label - the correct category name attached to a training example

classifier - a model that assigns an input to one of several categories

confidence - the model's numerical estimate of how strongly an input matches a class; it is not a guarantee of correctness

generalize - to work correctly on new examples that were not in the training set

debugging - finding and fixing a problem in a program or system

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus the difference between Thursday's Scratch project and Teachable Machine. Start from what he actually did: in Scratch, he wrote an IF/ELSE rule using temperature; in Teachable Machine, he provided labeled images of cup, book, and pencil and the system learned patterns. Define rule-based program, machine learning, training data, label, classifier, confidence, generalize, and debugging. Give two examples: a thermostat with an explicit threshold is rule-based; an image classifier trained from many labeled pictures is machine learning. Emphasize that 99% confidence does not mean "99% guaranteed correct"; confidence describes the model's internal certainty among the classes it knows. Ask: "Who wrote the Scratch decision rule?" and "What did Teachable Machine use instead?" Then guide the online variation experiment. Do not collect faces or personal information. Let Atticus observe mistakes rather than trying to make the model look perfect.

WORKED EXAMPLE 1: Rule-based

Problem: IF temperature > 75 THEN say "warm" ELSE say "not above 75."

Solution: The programmer wrote the decision boundary.

Why it works: The computer follows the rule exactly; it does not learn it from examples.

WORKED EXAMPLE 2: Machine learning

Problem: Train CUP, BOOK, and PENCIL from labeled images.

Solution: The model learns visual patterns from the training examples.

Why it works: The programmer supplies data and labels, but not a complete hand-written rule for every possible image.

WORKED EXAMPLE 3: Confidence is not certainty

Problem: Model says "cup: 98%" for an unfamiliar object.

Solution: The prediction may still be wrong.

Why it works: The model chooses among patterns/classes it learned; high confidence does not prove the world actually matches the prediction.

OPEN

<https://teachablemachine.withgoogle.com/>

Mission: Test whether your classifier works on NEW versions of the same objects

1. Reopen or rebuild CUP, BOOK, and PENCIL classes. If yesterday's model is gone, collect about 15-20 examples per class.
2. Train the model.
3. Test each object in the SAME position/background used during training. Record confidence.
4. Change ONE condition: rotate the object, move farther away, change the background, or change lighting. Test again.
5. Record whether the class stayed correct and how confidence changed.
6. Find the class with the weakest or least stable result. Add 10-15 MORE varied training examples for that class.
7. Retrain. Test the difficult condition again. Did the result improve?
8. Try one completely unfamiliar household object. Record what class the model chooses and why that does NOT prove the object really belongs to that class.

Test	Condition	Prediction	Confidence	Correct?

1. What changed when you tested a new angle/background/light condition?

2. Did adding more varied training examples improve the weakest class? Explain with evidence.

3. What did the model do with an unfamiliar object?

1. Who wrote the decision rule in Thursday's Scratch program?

2. What did Teachable Machine use to learn its categories?

3. What is one important difference between a rule-based program and a machine-learning classifier?

4. Why can a classifier be highly confident and still be wrong?

5. What does “generalize” mean in machine learning?

6. What does debugging mean in your own words?

WORDS YOU NEED TODAY

visual hierarchy - the order in which a viewer notices information

contrast - a noticeable difference that helps important elements stand out

symbol - a simple image or mark that represents an idea

annotation - a short label or note added to explain part of a visual

composition - the arrangement of elements on a page

caption - brief text that explains an image or diagram

LISTEN FIRST - VOICE TUTOR PROMPT

Act as an art/design teacher. Explain visual hierarchy, contrast, symbol, annotation, composition, and caption using simple examples. Atticus will create a one-page visual explainer of ONE concept from Week 1: decimal division, bridge expansion, light transmission, the Greek polis, French negation, or rule-based vs machine learning. He should not simply decorate the page. Help him plan what a viewer should notice first, second, and third. Require a clear title, one central diagram or visual, at least three useful labels/annotations, and a 2-3 sentence caption that teaches the concept accurately. Do not design the final piece for him; ask questions that help him choose the best layout.

CHOOSE ONE TOPIC

- Decimal division
- Why bridges need expansion joints
- Transparent vs translucent vs opaque
- How geography shaped the Greek polis
- French ne...pas and elision
- Rule-based programs vs machine learning

REQUIREMENTS

- Clear title
- One main diagram/visual
- At least 3 labels or annotations
- One deliberate use of contrast or size for hierarchy
- 2-3 sentence caption that accurately teaches the idea
- Your name + date

PLAN IT HERE - CREATE THE FINAL ON A SEPARATE BLANK SHEET

What should the viewer notice first? _____

Main visual/diagram: _____

Three labels you will include: _____

Caption idea: _____

What I can do now

• One math skill I can now do more confidently:

• One reading/writing skill I improved:

• One science idea I can explain:

• One history idea I can explain:

• One French sentence pattern I can use:

• One thing I learned about programming or AI:

• One question I still have:

NOTHING MISSING CHECK

- Math independent practice + exit ticket
- Literature questions + summary
- Writing paragraph
- Science lab + explanation
- History questions
- French practice + oral check
- CS/AI online lab + exit ticket
- Art/design visual explainer

Mathematics

quotient: answer to division.

dividend: number being divided.

divisor: number you divide by.

equivalent: equal in value.

estimate: close prediction.

magnitude: size of a quantity.

Literature/Writing

thermal expansion: increase in size as material warms.

contract: become smaller.

expansion joint: gap/flexible connection that allows movement.

stress: internal force in material.

central idea: author's main point.

objective summary: main ideas without opinion.

reasoning: why evidence supports the explanation.

transition: word/phrase that connects ideas.

revise: improve writing.

Science

transmit: let light pass through.

scatter: redirect light in many directions.

transparent: passes most light with little scattering.

translucent: passes some light but scatters it.

opaque: does not transmit visible light.

absorb: take in light energy.

History

polis: independent Greek city-state/community.

politically independent: self-governing.

culturally connected: sharing language/beliefs/customs.

harbor: sheltered place for ships.

trade: exchange of goods/resources.

French

subject pronoun: word telling who does the action.

stem: verb part left after removing -er.

ending: letters added to match subject.

conjugate: change verb to match subject.

negation: make a sentence negative.

elision: drop a vowel before another vowel sound.

Computer Science/AI

rule-based program: follows explicit programmer-written rules.

machine learning: learns patterns from examples.

training data: examples used to train model.

label: correct category name.

classifier: model that chooses a category.

confidence: model's strength of belief, not a guarantee.

generalize: work on new examples.

debugging: find and fix problems.

Art/Design

visual hierarchy: what the viewer notices first/next.

contrast: difference that creates emphasis.

symbol: simple mark representing an idea.

annotation: explanatory label.

composition: arrangement on the page.

caption: brief text explaining a visual.

THURSDAY MASTERY DECISIONS USED TO BUILD FRIDAY

- Math: fraction-division procedure is now proficient/near mastery based on repeated correct independent work; conceptual explanation still develops. Decimal multiplication is proficient but estimation is inconsistent. Friday moves to decimal division with an estimation repair.
- Literature: comprehension is solid; precision of topic and objective summary remains developing.
- Science: light-path reasoning is proficient; transparent/translucent/opaque vocabulary needs explicit repair.
- History: geography -> city-state causal reasoning is strong; role of sea and culturally connected vs politically unified need reinforcement.
- French: ne...pas is mostly secure; elision and subject/verb endings need attention.
- CS/AI: practical online work completed; conceptual comparison unfinished, so Friday focuses on rule-based vs ML and generalization.

Math answers

Guided: 1) 2.4 2) 7 3) 42 4) 4.5 5) 14 pieces.

Independent: 1) 12 2) 36 3) 52 4) 4.2 5) 12 packages 6) \$6.30.

Exit: E1) 7 E2) 27 E3) 25 E4) multiplying both dividend and divisor by 100 creates an equivalent ratio/division problem, so the quotient is unchanged E5) estimate about 9; exact 9.

Friday math decision rule

If decimal division exit ticket is $\frac{4}{5}$ or better AND Atticus can explain E4 in his own words, continue Monday into ratios/equivalent ratios while keeping short decimal fluency warmups. If procedure is strong but E4 is weak, continue but begin Monday with a 5-minute conceptual repair. If below $\frac{4}{5}$, reteach decimal division before ratios.

Literature - expected ideas

1. Topic: bridge movement / thermal expansion in bridges / bridge expansion and design.
2. Central idea: Bridges change size slightly with temperature, so engineers design joints/bearings that allow safe controlled movement and prevent damaging stress.
3. Many materials expand when warmed because particle motion increases and the material occupies slightly more space.
4. Stress: internal force/pressure in a material when it is pushed, pulled, bent, or prevented from moving.
5. Blocking expansion causes internal stress that can contribute to cracking, bending, or connection damage.
6. Expansion joints and bearings (or equivalent described design features).
7. The ruler shows that small size changes are hard to notice in short objects but add up across a long bridge.
8. Strong structures may be designed to move in controlled ways rather than being completely rigid.
9. Summary should include temperature-driven movement, damage risk when blocked, and engineering features that allow safe movement.

Writing rubric - 20 points

- 4 - clear main explanation answers why movement is allowed
- 4 - at least two accurate pieces of article evidence
- 4 - reasoning connects evidence to design
- 4 - organization / transitions / conclusion
- 4 - sentence boundaries, capitalization, punctuation, spelling

Science - expected ideas

Transparent: high transmission + clear image; translucent: some transmission + scattering / blurry image; opaque: no meaningful visible-light transmission through material.

Q2 should explicitly state that light can pass even when scattering prevents a clear image.

CER on wax paper: some light is transmitted, but scattering redirects rays so detailed image information is lost.

Challenge: no; classification depends on the balance of transmission/scattering, not simply whether any reflection and transmission occur.

History - expected ideas

1. Polis: an independent Greek city-state/community including its central settlement and surrounding territory.
2. Mountains made land travel difficult and separated communities, encouraging independent governments.
3. Any three of trade, transportation, fishing, communication, movement of people/goods/ideas.
4. Trade supplied food/resources that limited farmland could not produce easily.
5. Culturally connected = shared language/beliefs/customs; politically unified = ruled by one government.
6. mountains -> difficult land travel -> separated communities -> independent political development/city-states.
7. Shared culture did not erase competition over land, trade, power, or different local interests.

French

A1 Je n'aime pas les maths. A2 Tu n'écoutes pas la radio. A3 Elle ne regarde pas un film. A4 Nous ne parlons pas français. A5 Ils ne jouent pas au tennis.

B6 joue B7 aimes B8 regardons B9 écoute B10 parlent.

C answers vary; require correct subject/verb agreement and correct ne/n'...pas if negative.

Computer Science / AI

1. Atticus/the programmer wrote the Scratch rule explicitly.
2. Teachable Machine used labeled training examples/images.
3. Rule-based systems follow explicit rules; ML classifiers learn patterns from examples.
4. Confidence is the model's internal score among learned classes; limited/biased data or unfamiliar inputs can still produce confident mistakes.
5. Generalize = perform correctly on new examples/conditions not seen exactly during training.
6. Debugging = finding, understanding, and fixing a problem in a program/system.

Art / Design quick rubric - 10 points

2 clear hierarchy/title

2 accurate central visual

2 useful labels/annotations

2 caption teaches the concept accurately

2 craftsmanship / deliberate composition